

A Replication of Study 3 in Stellar et al. (2018): Awe and Humility

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Abstract As part of the CREP project... we follow a direct replication and extension approach.

The seminal findings of the Open Science Collaboration [1] triggered what is now known as the ‘replication crisis’ in psychological science, prompting researchers to question thusfar widely unchallenged publications by and large. Since then, there has been an ongoing effort within the psychological research community to actively pursue replication of affected studies with important findings, aiming to reevaluate their validity and ultimately restore trust in the body of psychological knowledge. As part of the Collaborative Replications and Education Project (CREP) project of J. E. Grahe *et al.* [2], we aim to contribute to the endeavour described above, investigating the replicability of the results in ‘Study 3’ of J. E. Stellar, A. Gordon, C. L. Anderson, P. K. Piff, G. D. McNeil, and D. Keltner [3]: Awe and Humility. In this study, the authors hypothesize momentary experiences of awe to cause a temporary increase in humility. The authors operationalize this hypothesis by means of a randomized video intervention, designed to induce either a state of awe or a neutral state, with a subsequent evaluation on a behavioral measure of humility. Both theoretical background and original study design are outlined in detail in the original publication of J. E. Stellar, A. Gordon, C. L. Anderson, P. K. Piff, G. D. McNeil, and D. Keltner [3]. To ensure transparency, following the general recommendation of adopting the principles of open science [4], we have made all relevant materials for this replication study available on the platform Open Science Framework (OSF), in particular the preregistration form at <https://osf.io/537aq>.

A Direct Replication and Extension of Stellar et al. (2018)

To ensure a direct replication, we aim to adopt the authors original methodology and analysis wherever possible. As our recruitment of participants has taken place in germany, however, we have opted to translate the study material into german, in particular the audio track for the neutral condition video. Our recruitment process follows the snowball sampling principle [5], consequently our sample consists of mostly students, as well as their friends and relatives, resulting in demographics distinct from to the original sample of J. E. Stellar, A. Gordon, C. L. Anderson, P. K. Piff, G. D. McNeil, and D. Keltner [3]. All known deviations are documented in their entirety within the preregistration form. For the direct replication, we aim to replicate the following hypothesis.

In addition to our direct replication, we introduce a third condition, in which participants are shown a more modern video, which is also designed to induce awe. Based on this condition, we extend our study the following confirmatory hypothesis.

Furthermore, we aim to investigate the natural follow-up question of how the two experimental conditions compare by means of the following exploratory hypothesis.

Note, that as documented within the preregistration, we have formulated a total of six supplementary exploratory hypotheses. Of these, E1 was chosen at random prior to any further data analyses. The full list of all exploratory hypotheses is documented in the publicly available preregistration form.

Method

The scripts required for reproducing the following results have been made publically available on GitHub at https://github.com/TiziSchuh/Stellar_Rep_01. The hypotheses, operationalizations, analysis plan, inference criteria, and sample size justification have been preregistered at <https://osf.io/537aq>. Open material and data can all be found at <https://osf.io/cy287>.

Participants

The initial sample had 189 participants. We preregistered to exclude participants who did not write anything in the writing assignment (26 exclusions) or indicated that their data should not be used (1 exclusions), leading to a final sample size of $n = 162$. The final sample had a mean age of 32 ($SD = 16.2$, range: 18–77). Neutral condition 50, experimental condition 58 and new experimental condition 54

Measures and Materials

Independent Variable (IV): Video condition (Neutral vs. Awe Original vs. Awe Modern). Dependent Variable (DV): State humility, operationalized as the log-transformed ratio of listed personal strengths to weaknesses. The number of strengths and weaknesses are counted by two independent raters from the free-text answer of the participants in a writing exercise. We use the “Codebook Instructions for Counting Participant-Reported Strengths and Weaknesses” from <https://doi.org/10.26443/msurj.v19i1.231> (Appendix B)

The awe emotional state is computed as the mean of three items (Item1: Staunen, Item 2: Faszination, Item 3: Ehrfurcht.)

- measure of awe (3 out of 5 items in 7? point likert scale)
- measure of humility (writing exercise, count and transform...)

Procedure

One-time, cross-sectional online study. Data collection: 19.06.2026 - 10.07.2026. Participants first watch the video, then do the strength/weakness writing exercise, then report on their emotional state. Finally they report on the control questions.

Statistical Procedure

Exclude persons with zero weakness to calculate log. Through log we get more normally distributed data. For H1 and H2, we perform t-test with a p-value of...

Deviations from Preregistration

Power not enough bc valid participants?

Results

Confirmatory Results

The results was: $\Delta M = 0.09$, 95% CI $[-0.10, \infty]$, $t(71) = 0.80$, $p = .212$.

Exploratory Results

The new video... (just as effective on awe measure, effect also non-significant)

Discussion

Limitations and Future Directions

lack of evidence is not evidence of absence. Maybe just power not big enough or unlucky... but for this: CREP. Also, we have not tested for MI, explicitly investigating to what extent the translation may have an impact on the reproducibility of the effect would adress this limitation.

Conclusion

References

This Section Is an Appendix

Another Appendix

Bibliography

- [1] Open Science Collaboration, "Estimating the reproducibility of psychological science," *Science*, vol. 349, no. 6251, p. aac4716, 2015.
- [2] J. E. Grahe *et al.*, "Collaborative Replications and Education Project (CREP)." [Online]. Available: <https://doi.org/10.17605/OSF.IO/WFC6U>
- [3] J. E. Stellar, A. Gordon, C. L. Anderson, P. K. Piff, G. D. McNeil, and D. Keltner, "Awe and humility," *Journal of personality and social psychology*, vol. 114, no. 2, p. 258, 2018.
- [4] F. D. Schönbrodt, M. Maier, M. Heene, and M. Bühner, "Forschungstransparenz als hohes wissenschaftliches Gut stärken," *Psychologische Rundschau*, 2018.
- [5] L. A. Goodman, "Snowball sampling," *The annals of mathematical statistics*, pp. 148–170, 1961.